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Finite-cycle sensitivity and phase-resolved amplification in effective loop quantum Bianchi IX cosmology

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Abstract

Effective loop quantum cosmology replaces the singular contraction of the Bianchi IX universe with bounce–recollapse cycles. We quantify the sensitivity to initial data of two effective quantizations, based on extrinsic-curvature (K) and connection (A) holonomies, using outcome labels and trajectory separations sampled at volume extrema and at specified phase markers. Sampling at these events removes the dependence on the time parametrization of the integration, although the finite-time distances still depend on an explicit choice of phase-space coordinates and norm. The directional scale factors are monitored along each trajectory, the integration stops at the first sub-Planckian crossing or numerical failure, and physical exits are reported separately from unresolved integrations. For an initial rescaling $\Lambda = 20$, the median logarithmic amplification from the first to the second recollapse is 3.81 and 3.94 decades in the K and A models, respectively, for pairs passing linearity and noise controls. The marker-defined bounce window contributes 0.15 and 0.14 decades. We also report the dependence on the initial scale, the phase thresholds, and stricter axis-length cutoffs, and we examine a six-cycle ensemble in extended precision. These are conditional finite-time measurements: they provide evidence of strong recurrent sensitivity, but do not establish an asymptotic fractal basin boundary, a universal mixing rate, or preservation of information in the full quantum state. The results show that the amplification is concentrated outside the chosen bounce window, consistent with earlier observations of strong sensitivity near recollapse.

1 Introduction

The Belinskii–Khalatnikov–Lifshitz (BKL) conjecture motivates a description of the approach to a generic spacelike singularity by approximately local sequences of Kasner epochs, interrupted by curvature transitions [1]. The homogeneous Bianchi IX, or Mixmaster, universe is an important prototype [2–5]. Its classical chaotic dynamics can be studied through the uncertainty exponent of outcome boundaries [6]. Time parametrization requires care: numerical Lyapunov exponents are not invariant under arbitrary changes of clock, although the characterization of chaos is invariant under transformations satisfying appropriate hypotheses [7, 8]. Comparisons made at geometrically defined events therefore offer a useful complementary diagnostic.

In effective loop quantum cosmology (LQC), a quantum-geometric bounce replaces the singular contraction in many homogeneous models [9, 10]. In closed Bianchi IX models, bounces can be followed by classical recollapses. Early studies involving inverse-volume effects suggested suppression of Mixmaster chaos [11, 12], whereas later work emphasized sensitivity accumulated across repeated classical segments [13]. Blackmore and Wilson-Ewing numerically found nearby solutions

separating especially strongly near recollapse and comparatively little at the bounce [14]. This qualitative mechanism is therefore already established, and it motivates the present measurements.

Different quantum treatments need not agree on this question. Deformed-algebra and generalized-uncertainty approaches [15, 16], quantum moments [17], Wheeler–DeWitt wave packets [18], and volume polymerization [19] modify different parts of the dynamics. We restrict attention to two specified effective holonomy Hamiltonians without inverse-triad corrections. We do not test the full quantum dynamics or attempt to decide between these models.

Recent work quantified the incomplete development of classical mixing before a Planck-scale curvature cutoff [20]. This raises a related finite-time question: how much sensitivity accumulates in the separate portions of a bouncing cycle? Our contribution is an ensemble measurement based on a common event construction, a comparison of the K and A quantizations, pathwise validity checks, and explicit numerical and statistical controls. Throughout, we distinguish finite-cycle amplification from claims about asymptotic chaos or information-theoretic mixing.

2 Effective LQC of Bianchi IX

We consider the diagonal vacuum Bianchi IX metric $ds^2 = -N^2dt^2 + \sum_i a_i^2\sigma_i^2$, with σ_i the left-invariant one-forms on S^3, and use units $G = \hbar = 1$. In terms of the densitized-triad variables $p_1 = a_2a_3$, $p_2 = a_1a_3$, $p_3 = a_1a_2$ and their conjugates K_i, $\{K_i, p_j\} = 8\pi\delta_{ij}$, the Hamiltonian constraint with lapse $N = \sqrt{q}$, $q = p_1p_2p_3$, reads

$$\mathcal{C} = -\frac{1}{8\pi}\sum_{i<j} p_ip_jK_iK_j + U, \tag{1}$$
$$U = \frac{1}{32\pi}\Big[\sum_i \Big(\frac{p_jp_k}{p_i}\Big)^2 - 2\sum_i p_i^2\Big],$$

where (i, j, k) is a cyclic permutation and ${}^3R = -16\pi U/q$. We verified equation (1) against the curvature of the Bianchi IX metric computed directly in Euler coordinates (appendix A).

The first effective model, which we call the K quantization, follows references [14, 21]. Holonomies of the extrinsic curvature replace K_i by $\sin\theta_i/(\gamma\bar{\mu}_i)$, with $\theta_i = \gamma\bar{\mu}_iK_i$, $\bar{\mu}_i = \sqrt{\Delta p_i/(p_jp_k)}$, $\Delta = 4\sqrt{3}\pi\gamma$ and $\gamma = 0.2375$, so that

$$\mathcal{C}_K = -\frac{q}{8\pi\gamma^2\Delta}\sum_{i<j}\sin\theta_i\sin\theta_j + U. \tag{2}$$

The second model, the A quantization, is Wilson-Ewing's effective constraint [10], built from holonomies of the Ashtekar–Barbero connection $c_i = \gamma K_i + \Gamma_i(p)$, where Γ_i is the spin connection. Because $\Gamma_i = \partial_{p_i}F(p)$ for some F, the map $(c, p) \to (K, p)$ is canonical, and we find that for $\ell_o = 1$ Wilson-Ewing's classical constraint coincides *exactly* with equation (1). The two quantizations therefore share the same classical limit and differ only by quantum corrections. Written in the present variables, equation (4.2) of reference [10] (with $r_o = 2$) reads

$$\mathcal{C}_A = -\frac{q}{8\pi\gamma^2\Delta}\sum_{i<j} s_is_j + \frac{1+\gamma^2}{\gamma^2}U - \frac{1}{8\pi\gamma^2\sqrt{\Delta}}\sum_k \frac{(p_ip_j)^{3/2}}{\sqrt{p_k}}\,s_k, \tag{3}$$

with

$$s_i = \sin(\theta_i + \phi_i), \qquad \phi_1 = \frac{\sqrt{\Delta}\,(a_1^2 - a_2^2 - a_3^2)}{2a_1a_2a_3} \quad \text{(and cyclic)}$$

where in the last sum of equation (3) (i, j) are the two indices different from k. Compared with equation (2), the holonomies are shifted by the spin-connection phases ϕ_i, a term linear in s_k appears, and the potential is rescaled; the classical limit of all three modifications is exactly equation (1). The spin-connection phases depend on anisotropy as well as on overall size, so the magnitude of the difference between the two quantizations cannot be inferred from the smallest axis alone.

For the numerical evolution we use the logarithmic triad variables $x_i = \ln p_i$ together with the angles θ_i, and choose the constant lapse $N = \gamma\sqrt{\Delta}$, which equals proper time up to a constant factor. For the K quantization the equations of motion then become

$$\begin{aligned} x_i' &= \cos\theta_i\,(\sin\theta_j + \sin\theta_k), \\ \theta_i' &= -S + \tfrac{1}{2}\sum_{l\neq i} x_l'(\theta_l - \theta_i) + F_i(x), \end{aligned} \tag{4}$$

with $S = \sum_{i<j} \sin\theta_i \sin\theta_j$ and the constraint $S = W(x)$, where W and F_i are the rescaled potential and its force (appendix A); the A quantization has the same structure. The angles enter only through sines, cosines and differences, so $\theta_i \to \theta_i + \pi$ (K) or $+2\pi$ (A), applied to all three angles at once, is an exact symmetry. We exploit this by storing $\theta_i = n_i\pi + \delta_i$ with integer n_i and $|\delta_i| \lesssim \pi/2$, so that $\sin\theta_i = (-1)^{n_i}\sin\delta_i$ and the differences $\theta_l - \theta_i$ are evaluated without cancellation. This reduces cancellation near the periodic branches, although it does not remove the need for extended precision when following small separations over many cycles.

An effective Hamiltonian assumes sufficiently sharply peaked states and neglects corrections that can matter at small directional lengths, so its domain of validity must be monitored along every trajectory. A cutoff $a_i > \ell_{\rm Pl}$ is only a minimal exclusion criterion; passing it does not prove the validity of the effective approximation. We locate downward crossings of $\min_i a_i = \ell_{\rm Pl}$ throughout every integration segment, including contractions, expansions, and multiple bounces, and terminate at the first crossing. Checking only a volume minimum is insufficient because a directional minimum can occur at another time. We also track crossings of $3\ell_{\rm Pl}$ and $10\ell_{\rm Pl}$ for stricter conditional analyses. Numerical exits are recorded separately when the normalized Hamiltonian residual exceeds 10^{-3}, a stage exhausts its integration budget, or another integration failure occurs. No trajectory is readmitted after an exit. State spreading and omitted inverse-triad effects remain additional limitations even for trajectories passing these cuts.

3 Ensembles and event-based measurements

Initial data are taken on the Cornish–Levin slice [6]: $\ln a_1 = 0$, $\ln a_2 = \Sigma/(1+v+uv)$, and $\ln a_3 = \Sigma(v+uv)/(1+v+uv)$, with Kasner-parametrized velocities and Σ fixed by the classical constraint. The contracting branch is rescaled by $a_i \mapsto e^{\Lambda} a_i$, and the third momentum is adjusted to satisfy the chosen effective constraint. We sample (u, v) uniformly in $[1, 2]^2$, with reproducible independent point seeds and uniformly distributed perturbation directions. The two quantizations share the same sampled points and directions.

The parameter Λ specifies the initial size; it is not itself an observed e-fold count. The actual contraction depth before the first bounce B_1 is measured as

$$\begin{aligned} N_V &= \ln(V_{\rm initial}/V_{B_1}), \\ N_a &= \ln(a_{\rm initial}/a_{B_1}) = N_V/3, \end{aligned} \tag{5}$$

Table 1. Measured initial contraction depth. Λ is an initial rescaling parameter.

model	Λ	N_V	N_a
K	10	11.48	3.83
K	20	21.94	7.31
K	30	32.41	10.80
A	10	11.24	3.75
A	20	21.72	7.24
A	30	32.25	10.75

where $a = V^{1/3}$. We use $\Lambda = 10, 20, 30$. The bound of about 30 e-folds discussed in reference [20] refers to N_a, not N_V, and to specified cosmological and astrophysical circumstances, so our scan does not test the full range of that bound. Table 1 gives the measured median depths among base trajectories reaching B_1.

Each cycle is divided into phases by geometric events defined through the phase marker $h = \max_i |\sin\theta_i|$. The onset On is the first upward crossing of $h_0 = 0.5$ while contracting, and B is the subsequent first volume minimum. The end of the quantum phase, Off, is the last downward crossing of h_0 before h reaches 0.05 without re-entry; a look-ahead integration confirms this event. The recollapse R is the subsequent volume maximum with $h < 0.01$, skipping maxima associated with further quantum oscillations, and P is the point where $\ln V$ has decreased by five after R. All segments remain subject to the same cumulative validity monitor. Multiple crossings are allowed rather than silently identified with a single bounce, and the phase analysis is repeated with $h_0 = 0.3$ and 0.7.

The marker h is a convention for locating a bounce window, not a directional Hubble rate or a general criterion of classicality. In particular, the A dynamics involve the shifted angles $\theta_i + \phi_i$ and additional spin-connection terms. Event locations and numerical amplification factors depend on this convention, which motivates the threshold comparison. Our results do not assume that a small value of h alone implies that every quantum correction is negligible.

Sensitivity is measured through the separation of nearby trajectories. For initial points separated by ε in the (u, v) plane, we define

$$d_E(\varepsilon) = \left[\sum_i (\ln a_i^{\rm pert}(E) - \ln a_i^{\rm base}(E))^2\right]^{1/2} . \tag{6}$$

The phase gain is $g_{E\to F} = \log_{10}(d_F/d_E)$. Each trajectory is sampled at its own occurrence of the event. These quantities are invariant under any monotonically increasing relabeling of the integration time, but they depend on the coordinates, the projection onto scale factors, and the norm. We do not identify them with a full phase-space entropy or an invariant asymptotic Lyapunov exponent.

The phase ensemble has 1000 base points for each depth and quantization. Each point is perturbed with $\varepsilon = 10^{-9}$ and 10^{-10} and is accompanied by an unperturbed control integrated with a tolerance ten times tighter. The complete recollapse cycle $R_1 \to R_2$ is split as $R_1 \to P_1 \to {\rm On}_2 \to {\rm Off}_2 \to R_2$. A common cohort is used for all four phase gains and their total. At every endpoint, a pair must remain above the chosen length cutoff, have $d_E(10^{-10}) \le 0.1$, have numerical control separation at most $0.01 d_E$, and satisfy

$$\frac{\|\Delta \ln a_E(10^{-10}) - 0.1\Delta \ln a_E(10^{-9})\|}{d_E(10^{-10})} \le 0.1. \tag{7}$$

We report the number of eligible pairs before these controls and of resolved pairs

after them. Selection by numerical resolvability can itself affect a conditional statistic and does not guarantee unbiased sampling of the population.

For each retained pair the four logarithmic gains telescope to the full-cycle gain. We compute this total for each pair before taking the median, and never replace it by the sum of the marginal phase medians. Intervals in the tables and figures are 95% percentile-bootstrap intervals of the median using 2000 resamples of independent base points. They describe sampling uncertainty within the retained cohort, not model uncertainty or the effect of excluded trajectories.

Outcome sensitivity is measured separately through discrete labels. We record the smallest axis at B and R and the fastest-contracting axis at On. For 400 base points per quantization at $\Lambda = 20$, we measure the fraction $f(\varepsilon)$ of differing labels at 17 perturbation radii from 10^{-2} to 10^{-10}. Restricting base points to $[1.01, 1.99]^2$ keeps all perturbed points inside the initial square. At each radius, only pairs in which both trajectories reach the event before an exit are counted, and the resulting denominator is recorded. Wilson binomial intervals accompany the fractions; repeated radii share base points and are correlated. Zeros are reported as zero observed counts with upper confidence bounds, not as exactly zero probability. Where the level $f = 0.1$ is bracketed by the sampled radii, we report that bracket rather than inserting an arbitrary probability floor or asserting a precisely measured smoothing scale. Unequal label probabilities need not produce the independent-label plateau 2/3; the corresponding expression is $1 - \sum_i p_i^2$.

To test whether the growth persists over longer evolutions, we integrate 64 base points per quantization over six cycles in double-double arithmetic, with perturbations 10^{-24} and 10^{-25}, relative tolerance 10^{-28}, and an unperturbed control at 10^{-29}. All scale-factor logarithms are stored as high and low components in 64-bit floating point (Float64) so that small separations are preserved. The same numerical quality requirements apply, with the perturbation ratio in equation (7) unchanged. Successive-event gains use pairs resolved at both ends; a separate six-cycle mean uses one cohort resolved at every event of the chosen type. We report attrition and do not infer properties of trajectories beyond their first exit.

Finally, the original classical calculation is retained as a diagnostic reference. With the Cornish–Levin first-$u > 7$ outcome, an integration-depth cutoff $|\ln V| = 10^{13}$, and 2000 initial points, the uncertainty fit gives $D = 1.834 \pm 0.004$. The quoted fit error excludes depth, unresolved-outcome, and fit-window systematics. The value is close to, but not consistent within these stated errors with, the published 1.86 ± 0.01 [6]; scale-window fits range approximately from 1.81 to 1.86. This is an approximate classical reproduction, not an LQC dimension measurement. The relation $f \propto \varepsilon^{2-D}$ requires an appropriate limiting boundary and sampling measure [22, 23].

4 Results

4.1 Finite-event outcome maps

Figures 1 and 2 describe outcome sensitivity at individual numbered events. The increasingly interleaved labels motivate a quantitative measure of this refinement, but do not establish a noninteger asymptotic boundary dimension in LQC. A finite sample with no label changes at a small radius supplies an upper bound on the disagreement probability. We therefore do not extrapolate a finite-cycle crossover to infinitely fine structure.

Table 2 gives the sampled crossover brackets. They describe finite-event sensitivity and the precision of the sampled grid, not uncertainty intervals for an

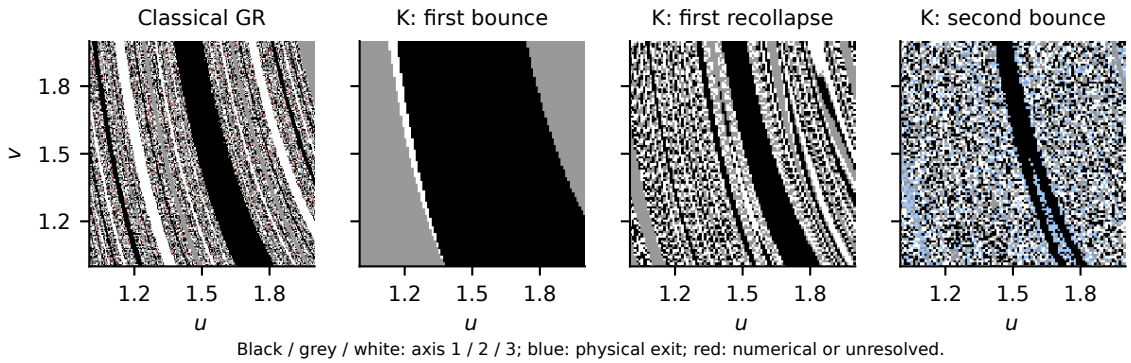


Figure 1. Classical reference map (left, 200^2 points) and corrected finite-event K maps at $\Lambda = 20$ (100^2 points). Classical labels use the first-$u > 7$ outcome; LQC labels identify the smallest axis at the indicated event. Blue indicates a physical exit before the event; red indicates a numerical failure or unresolved event. An outcome is retained only if the path up to that event passes the minimal cutoff. Later failures do not retrospectively remove earlier valid events.

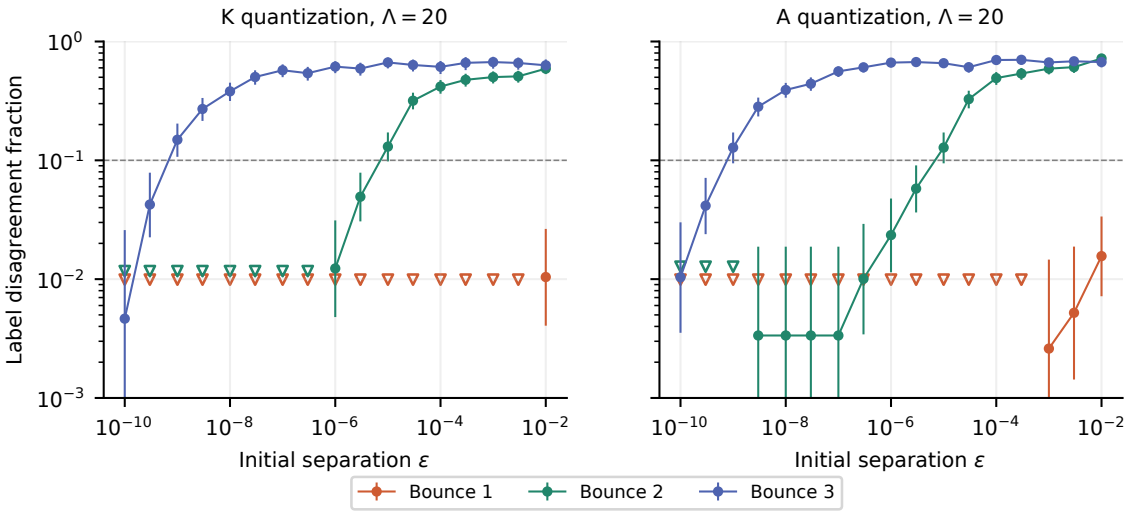


Figure 2. Label disagreement at the first three bounces, with 95% Wilson intervals. Open downward triangles denote the upper bounds when zero disagreements are observed; their vertical coordinates are not measured nonzero fractions. Each radius has its own recorded valid-pair denominator. The dashed line marks $f = 0.1$.

asymptotic exponent. Nonzero label changes in the tighter-tolerance control limit the interpretation of small measured fractions at late events. Conversely, the absence of observed control flips only bounds their probability; it does not certify every label at arbitrarily small perturbations.

4.2 Amplification over a resolved cycle

At $\Lambda = 20$ the measured bounce-window gains are 0.15 (K) and 0.14 (A) decades, compared with 3.81 and 3.94 for the full cycle (table 3). Figure 3 shows the individual phase gains for the same cohort. The phase decomposition quantifies amplification outside the selected bounce window, including the turnaround and subsequent contraction. A large amplification per unit of contraction near recollapse does not imply that most of the total is generated in a fixed five-e-fold window. The relative contributions depend on depth and on the chosen partition.

Figure 4 compares initial scales without identifying Λ with an exact contraction depth. Differences between the K and A medians should be interpreted in light of their sampling intervals and their different retained populations. Similar finite-time

Table 2. Sampled brackets for the level $f = 0.1$ and label changes in the tighter-tolerance control. Unbracketed levels are not assigned an artificial smoothing scale.

model	event	bracket in ε	noise flips/pairs
K	B_1	unbracketed	0/384
K	B_2	$[3 \times 10^{-6}, 1 \times 10^{-5}]$	0/322
K	B_3	$[3 \times 10^{-10}, 1 \times 10^{-9}]$	0/212
K	R_1	$[3 \times 10^{-4}, 1 \times 10^{-3}]$	0/381
K	R_2	$[3 \times 10^{-8}, 1 \times 10^{-7}]$	0/272
K	R_3	unbracketed	2/184
A	B_1	unbracketed	0/384
A	B_2	$[3 \times 10^{-6}, 1 \times 10^{-5}]$	0/298
A	B_3	$[3 \times 10^{-10}, 1 \times 10^{-9}]$	1/286
A	R_1	$[3 \times 10^{-4}, 1 \times 10^{-3}]$	0/384
A	R_2	$[3 \times 10^{-8}, 1 \times 10^{-7}]$	1/286
A	R_3	unbracketed	3/286

Table 3. Corrected phase and total gains from R_1 to R_2, in decades, using one common cohort. Counts are resolved/eligible pairs; brackets are 95% bootstrap confidence intervals of the median. The total is the median of the per-pair total, not a sum of phase medians.

model	Λ	pairs	bounce window	complete cycle
K	10	466/576	0.26 [0.25, 0.27]	3.31 [3.22, 3.38]
K	20	503/703	0.15 [0.14, 0.15]	3.81 [3.74, 3.89]
K	30	378/757	0.10 [0.09, 0.10]	4.24 [4.14, 4.34]
A	10	388/602	0.25 [0.24, 0.27]	3.49 [3.42, 3.59]
A	20	462/737	0.14 [0.14, 0.15]	3.94 [3.84, 4.00]
A	30	333/792	0.11 [0.10, 0.11]	4.39 [4.25, 4.47]

gains do not establish a universal rate.

4.3 Cutoff sensitivity, thresholds, and attrition

Table 6 separates the first-exit reasons. Table 4 tests whether the result depends on admitting axes only slightly above the Planck length. These are nested conditional populations; their medians need not coincide. The contrast between several decades of full-cycle gain and a small bounce-window gain persists under the $10\ell_{\mathrm{Pl}}$ cut. This robustness concerns the retained effective trajectories, not the validity of a sharply peaked quantum description for every such trajectory. Table 5 reports event survival without combining failed integrations with physical threshold crossings. The accompanying data list the first physical exits, constraint failures, exhausted budgets, and other failures separately. Because of attrition, the retained ensemble cannot be identified with a generic late-time invariant measure.

The threshold comparison in table 7 makes the conventional nature of the bounce-window gain explicit. It does not establish an exact boundary between classical and quantum evolution. Any statement that the bounce is neutral should therefore be read as a statement about the chosen observables, resolved pairs, and phase markers, rather than about all perturbation directions in the full state space.

4.4 Six-cycle sensitivity

The high-precision analysis uses two perturbation sizes and retains their high and low components at every cycle. It therefore involves neither a change of storage precision nor a loss of the small initial separations. Figure 5 and table 8 test persistence of finite-cycle growth; neither a small surviving cohort nor approximately similar increments proves an asymptotic limit. Table 9 makes the accompanying attrition explicit. Rejected quality-control pairs and first-exit counts remain part of the result; unobserved trajectories are not assumed to behave like the survivors. The nested cutoffs in table 8 also quantify how requiring a larger separation from the Planck length affects the already limited late-cycle cohorts.

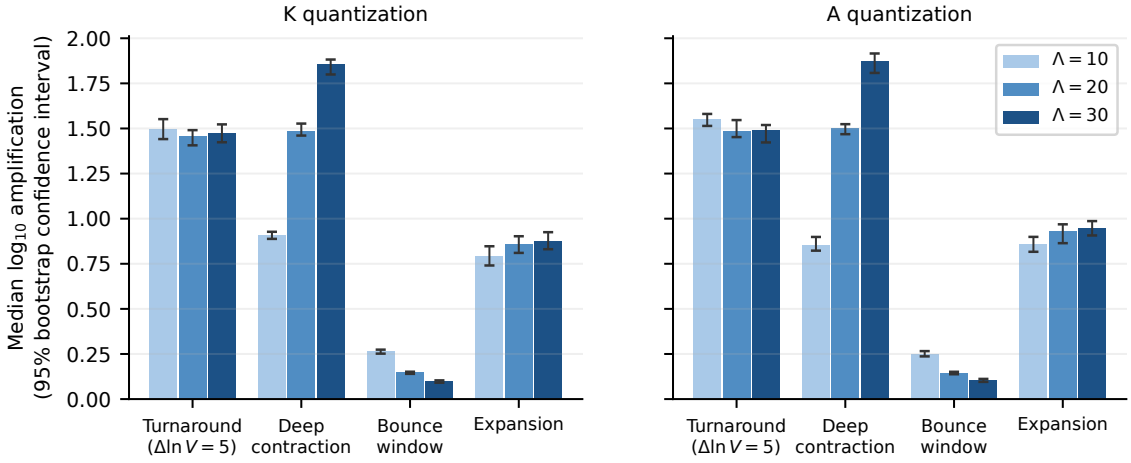


Figure 3. Phase gains for the same complete recollapse cycle and cohort as table 3. Phases are ordered from R_1 to R_2. The marginal phase medians need not add to the median total. Whiskers show 95% bootstrap intervals of medians, not the scatter of individual trajectories. The turnaround window is five volume e-folds, or 5/3 mean-scale-factor e-folds.

Table 4. Sensitivity to the minimum directional length over the entire preceding trajectory, at $\Lambda = 20$. The entry 1 is the minimal exclusion criterion, not a sufficient validity theorem. Counts and intervals have the same meaning as table 3.

model	cutoff / $\ell_{\rm Pl}$	pairs	bounce window	complete cycle
K	1	503/703	0.15 [0.14, 0.15]	3.81 [3.74, 3.89]
K	3	396/544	0.15 [0.14, 0.15]	3.94 [3.86, 4.03]
K	10	262/362	0.15 [0.14, 0.15]	4.01 [3.90, 4.08]
A	1	462/737	0.14 [0.14, 0.15]	3.94 [3.84, 4.00]
A	3	410/634	0.14 [0.14, 0.15]	3.96 [3.89, 4.08]
A	10	283/402	0.14 [0.13, 0.15]	3.97 [3.88, 4.15]

For the K quantization, only 9 of the 64 base trajectories complete all six cycles: 53 encounter the directional-length boundary and 2 exhaust an integration budget. The fixed resolved cohorts contain 8 bounce-sampled pairs and 7 recollapse-sampled pairs, with median per-cycle gains of 3.95 and 3.81 decades, respectively. The $3\ell_{\rm Pl}$ cut leaves only one resolved pair for each event type, and the $10\ell_{\rm Pl}$ cut leaves none. Thus the six-cycle K experiment cannot establish persistence well away from the length boundary. This limitation is distinct from the cutoff robustness of the first complete recollapse cycle in table 4. For the A quantization, 29 base trajectories complete six cycles, 30 reach the length boundary, and 5 exhaust the integration budget. At the minimal cut, 25 bounce-sampled and 14 recollapse-sampled pairs remain resolved at every respective event; their median per-cycle gains are 3.81 and 3.78 decades. Requiring all directional lengths to exceed $10\ell_{\rm Pl}$ reduces these cohorts to 5 and 2 pairs. Positive gains persist in these particular survivors, but such small cohorts do not support a precise ensemble comparison or a general late-cycle rate. Their bootstrap intervals reflect only resampling of the retained points and cannot account for the missing population.

5 Discussion and conclusions

The corrected analysis measures strong finite-cycle sensitivity in the two specified effective Hamiltonian models, with much smaller scale-factor amplification across the marker-defined bounce window than over a full recollapse cycle. This quantifies the earlier numerical observation reported in reference [14]. The event construction eliminates the dependence on an arbitrary integration clock, but not on the coordinate and norm choices underlying finite-time separations. The label maps

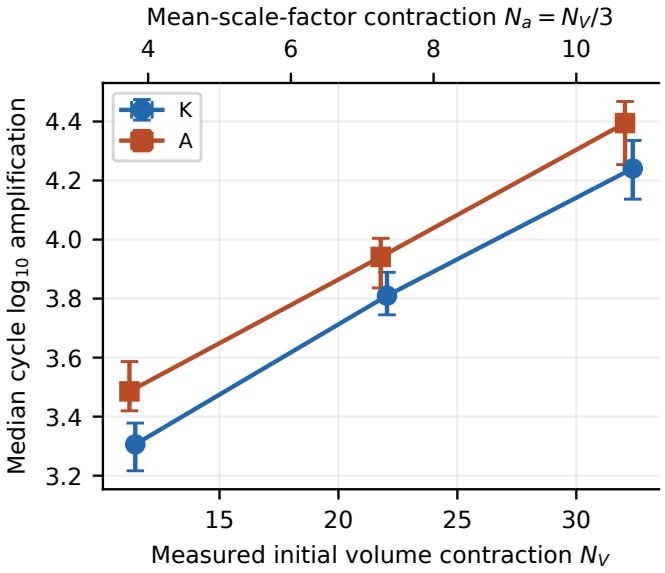


Figure 4. Median per-pair full-cycle gain versus measured median initial volume contraction, using the same resolved cohort for both summaries. The upper axis uses mean-scale-factor e-folds, differing by a factor of three. Both sets of bars are marginal 95% bootstrap intervals.

Table 5. Base trajectories reaching each recollapse before any physical or numerical exit, out of 1000 attempted points per row. Later exits do not invalidate an earlier event.

model	Λ	R_1	R_2	R_3
K	10	974	576	355
K	20	1000	714	500
K	30	993	770	610
A	10	1000	606	606
A	20	1000	742	742
A	30	1000	798	793

provide complementary outcome sensitivity, not an independently established fractal dimension of an asymptotic LQC boundary.

Three limitations are central. First, physical and numerical attrition condition every ensemble; the effective state need not remain sharply peaked even above the directional-length cutoff. Second, projected separations do not determine information transfer in the complete phase space or a quantum state. Third, a finite number of cycles cannot settle an infinite-cycle chaos or mixing question. A return-map study with tangent dynamics, suitable invariant-set evidence, and controlled treatment of escape from the valid regime would be needed for a stronger conclusion.

Our equations omit inverse-triad corrections [24] and do not test the same model as the early inverse-volume suppression arguments. We therefore make no general claim about whether quantum gravity creates or removes chaos. Matter, inhomogeneity, and quantum fluctuations can change the dynamics. Applications to black-hole interiors would require a separate model and cannot be inferred from recollapse measurements in this closed homogeneous cosmology.

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Table 6. First stop reason for the three-cycle phase runs (1000 base trajectories each), including the attempted segment $R_3 \to P_3$. Physical crossings, constraint failures, and budget exhaustion are distinct.

model	Λ	complete	physical	constraint	budget	other
K	10	355	645	0	0	0
K	20	500	500	0	0	0
K	30	610	387	0	3	0
A	10	606	372	0	22	0
A	20	742	246	0	12	0
A	30	793	188	1	15	3

Table 7. Phase-marker variation at $\Lambda = 20$. From the first 200 base points, one common cohort passes the validity and numerical controls at all endpoints for all three thresholds. Counts are identical within each model; this comparison holds the sampled points fixed.

model	h_0	resolved pairs	bounce window	complete cycle
K	0.3	114	0.19 [0.17, 0.21]	3.81 [3.65, 4.03]
K	0.5	114	0.15 [0.14, 0.16]	3.81 [3.65, 4.03]
K	0.7	114	0.12 [0.11, 0.13]	3.81 [3.65, 4.03]
A	0.3	105	0.20 [0.17, 0.23]	3.94 [3.74, 4.12]
A	0.5	105	0.16 [0.14, 0.18]	3.94 [3.74, 4.12]
A	0.7	105	0.13 [0.11, 0.15]	3.94 [3.74, 4.12]

Author contributions

Muhammad Ramadhan: conceptualization, methodology, software, formal analysis, investigation, validation, visualization, writing (original draft), and writing (review and editing).

Conflict of interest

The author declares no competing interests.

Ethics statement

This study is purely theoretical and computational and did not involve human participants, human data or tissue, or animals; ethical approval was therefore not required.

Data availability

The accompanying code-and-data package contains the equations, pinned numerical environments, raw trajectory records, deterministic seeds, controls, and scripts generating the tables and figures. The revised results are in `results/revised/`; the reproduction instructions distinguish them from the earlier exploratory outputs. A public repository and archival identifier have not yet been assigned to this revised package.

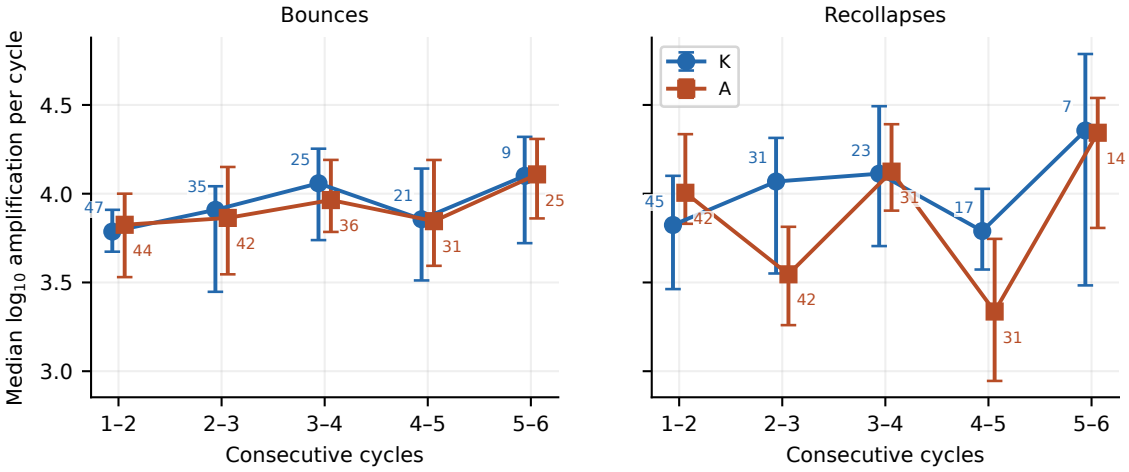


Figure 5. Successive-cycle gains in extended precision at the minimal directional-length cut, with 95% bootstrap intervals. Numbers next to points are resolved pair counts for that step. Different steps can have different retained cohorts; table 8 separately uses a fixed cohort across all six cycles.

Table 8. Mean gain per cycle from event 1 to event 6, calculated for each pair and then summarized over a fixed cohort resolved at all six events. B and R denote bounce and recollapse sampling. The length cutoff applies along the entire preceding trajectory; counts are resolved/eligible. An interval is not estimated for a single pair (CI n/a); small-cohort intervals require particular caution.

model	event	cutoff / ℓ_{Pl}	pairs	decades/cycle
K	B	1	8/9	3.95 [3.47, 4.08]
K	B	3	1/2	3.89 (CI n/a)
K	B	10	0/0	–
K	R	1	7/9	3.81 [3.41, 4.13]
K	R	3	1/1	3.75 (CI n/a)
K	R	10	0/0	–
A	B	1	25/30	3.81 [3.71, 4.02]
A	B	3	10/11	4.01 [3.93, 4.12]
A	B	10	5/5	4.12 [3.93, 4.27]
A	R	1	14/28	3.78 [3.58, 3.87]
A	R	3	4/11	4.04 [3.58, 4.20]
A	R	10	2/5	4.06 [3.92, 4.20]

Table 9. Base trajectories reaching each recollapse in the extended-precision experiment, out of 64 attempted points per model, before any physical or numerical exit. The last five columns separate completion from the first physical or numerical stop reason.

model	R_1	R_2	R_3	R_4	R_5	R_6	complete	physical	constraint	budget	other
K	63	46	32	24	18	9	9	53	0	2	0
A	64	49	49	38	38	29	29	30	0	5	0

A Explicit evolution equations and verification

Let $c = \gamma^2\Delta/4$ and, for cyclic (i, j, k), define

$$E_i = e^{x_j+x_k-3x_i}, \qquad e_i = e^{x_i-x_j-x_k}, \tag{8}$$

$$W = c\left(\sum_i E_i - 2\sum_i e_i\right), \tag{9}$$

$$F_i = c(-2E_i + 2E_j + 2E_k - 4e_i). \tag{10}$$

These specify equation (4). For the A quantization we set $\psi_i = \theta_i + \phi_i$, $s_i = \sin\psi_i$, $S_A = \sum_{i<j} s_i s_j$, $c_A = (1+\gamma^2)/\gamma^2$, and $\eta_i = \sqrt{\Delta}e^{(x_j+x_k-3x_i)/2}$. The equations are

$$x_i' = \cos\psi_i(s_j + s_k + \eta_i), \tag{11}$$

$$\theta_i' = -S_A + \tfrac{1}{2}\sum_{l\neq i} x_l'(\theta_l - \theta_i) - \sum_l x_l'\partial_{x_i}\phi_l - \sum_l \omega_{li}\eta_l s_l + c_A F_i, \tag{12}$$

where $\omega_{ii} = -1/2$ and $\omega_{li} = 3/2$ for $l \neq i$. The constraint is $S_A + \sum_l \eta_l s_l = c_A W$. With our constant lapse, the physical directional rates are $H_i = (\sum_j x_j'/2 - x_i')/(\gamma\sqrt{\Delta})$.

The verification scripts check the Bianchi IX curvature directly in Euler coordinates, the Arnowitt–Deser–Misner (ADM) Hamiltonian, the classical limits, canonical shifts, and the transformed Hamilton equations. The A implementation is compared with numerical derivatives of the connection Hamiltonian, and the classical Bianchi II transition and LQC Bianchi I bounce rules are reproduced. These checks establish algebraic and numerical consistency of the implemented models; they do not establish their physical applicability at every state or the statistical interpretation of a finite trajectory.

B Numerical controls and reproducibility

The integrator is a ninth-order Verner method with relative tolerance 10^{-12} for the phase and label runs and 10^{-28} for the extended-precision run. The absolute tolerances of the angles scale with their instantaneous magnitudes. Angles are rewrapped using their exact integer and residual representation. The onset search is staged through small marker thresholds, restarting its clock at each stage. This avoids loss of event-time resolution after a long preceding contraction. The root-based length guards use 20 interpolation checks per step, supplemented by accepted-step and endpoint monitoring. As with any numerical event search, resolution is finite; the regression counterexample is checked at two tolerances, and the statistical analysis includes independent tighter-tolerance controls. The extended-precision runs memoize the immutable Verner coefficient tables. Their values are checked byte for byte against the original constructors; complete controlled trajectory records were also verified to match exactly. This affects only the execution cost, not the integration algorithm or the accuracy settings.

The normalized constraint residual is $|S - W|/(\sum_i \sin^2\theta_i + c\sum_i(E_i + 2e_i))$ for the K quantization, and the analogous ratio with $s_i = \sin\psi_i$, the terms $|\eta_i s_i|$ and the factor c_A for the A quantization. The kinetic part is normalized by $\sum_i \sin^2\theta_i \geq |S|$, which vanishes only if all three sines vanish. Normalizing instead by $\sum_{i<j} |\sin\theta_i \sin\theta_j|$ fails at bounces where two sines pass through zero together: the denominator then collapses and absolute residuals $\sim 10^{-11}$ appear as relative errors of order unity. An earlier version of our analysis used that normalization and

Table 10. Sensitivity to numerical quality selection at $\Lambda = 20$ and the minimal length cut. Strict, primary, and relaxed selections require signal/noise at least 1000, 100, and 10, respectively, and relative vector discrepancies at most 0.05, 0.1, and 0.2. All retain the separation bound 0.1. The primary selection defines the headline results.

model	selection	resolved pairs	bounce window	complete cycle
K	strict	121	0.15 [0.13, 0.16]	3.94 [3.80, 4.06]
K	primary	503	0.15 [0.14, 0.15]	3.81 [3.74, 3.89]
K	relaxed	636	0.14 [0.14, 0.15]	3.82 [3.77, 3.90]
A	strict	104	0.15 [0.13, 0.16]	3.95 [3.74, 4.25]
A	primary	462	0.14 [0.14, 0.15]	3.94 [3.84, 4.00]
A	relaxed	642	0.15 [0.14, 0.15]	3.94 [3.87, 4.00]

terminated healthy trajectories; re-integrating a sample of them with the present definition gave no genuine constraint failure. A residual of 10^{-3} terminates the run; it is not an accuracy certificate for trajectory separations. Every event search has a budget of 300000 steps. Exhaustion of this budget is recorded as an unresolved integration, not as a physical exit. Perturbation and noise checks are made per pair at the events entering each statistic. Each trajectory has a cumulative cutoff mask; the mask follows all intermediate integrations, not only the volume extrema.

Table 10 shows that tightening the numerical controls substantially reduces the sample. The full-cycle medians remain several decades, while the bounce-window medians remain small. The shifts between these conditional populations are an additional selection sensitivity, not part of the bootstrap sampling intervals.

Files record every point, role, perturbation size, event, cutoff mask, stop reason, and code hash. The displayed initial-coordinate columns are Float64 summaries; exact extended-precision initial data are regenerated from the saved point ID and deterministic sampling rule. Resumption refuses mismatched source or parameter metadata. All revised production files use one schema, and the analysis refuses to generate manuscript artifacts until the declared runs are complete. The earlier mixed-precision six-cycle output is retained as historical data and is not used in the revised tables. Regression tests check that the previously missed sub-Planckian trajectory now terminates at the crossing, that a trajectory formerly stopped by the collapsing residual normalization completes its cycle with residual below 10^{-6} while a genuine constraint violation is still detected, and that the median total is distinguished from the sum of medians.

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